EliteDataScience

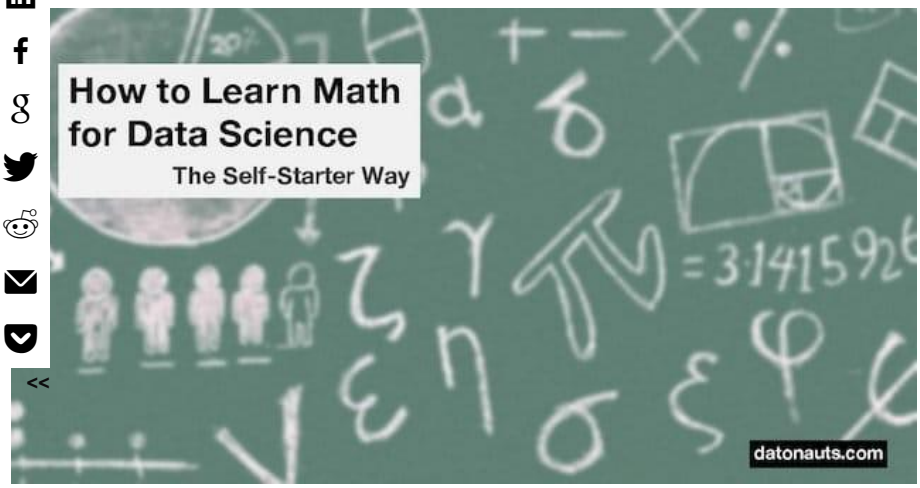
Free Data Science Courses Blog About

How to Learn Math for Data Science, The Self-Starter Way

October 30, 2016

Share Google Linkedin Tweet

Do you need to have a math Ph.D to become a data scientist? Absolutely not! This guide will show you how to learn math for data science and machine learning without taking slow, expensive courses.



How much math you'll do on a daily basis as a data scientist varies a lot depending on your role. Keep reading to find out which concepts you'll need to master to succeed for your goals.

Pre-requisite: Basic Python Skills

To complete this guide, you'll need at least basic Python* programming skills. We'll be learning math in an applied, hands-on way.

Check out our guide, How to Learn Python for Data Science, The Self-Starter Way, for the fastest way to get up to speed with Python. We recommend at least completing up to **Step 2** in that guide.

**note: other languages are fine too, but the examples will be in Python.*

Math Needed for Data Science

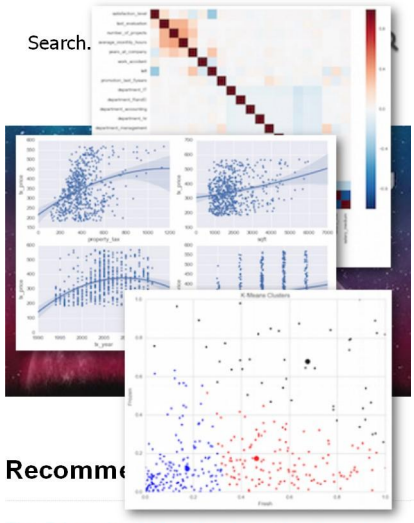
The amount of math you'll need depends on the role. First, every data scientist needs to know some statistics and probability theory. We have a guide for that:

- How to Learn Statistics for Data Science, The Self-Starter Way

What about other types of math? Well, here's where the answer is more nuanced... it

Search.

How to Learn Practical Machine Learning in Just 1 Month...



Recomme

The 5 Levels of Machine Learning Iteration

Skip the academics. Learn practical ML from practicing professionals who will teach you **how to get real results with machine learning.**

R vs Python for Data Science: Summary of Modern Advances

Python Machine Learning Tutorial, Scikit-Learn: Wine Snob Edition

Learn more about this 100% project-based machine learning masterclass

Keras Tutorial: The Ultimate Beginner's Guide to Deep Learning in Python

21 Must-Know Machine Learning Interview Questions and Answers

Supercharge Your Data Science Career!

Download Our Hand-Picked List of 88 Awesome Free Resources

Enter your First Name...

Enter your Email Address...

GET INSTANT ACCESS

We'll never sell or spam your email.

depends on how much *original machine learning research* you'll be doing.

How to Learn Practical Machine Learning in Just 1 Month

Application-Heavy Machine Learning Positions

In practice, especially in entry-level roles, you'll often be using out-of-the-box ML implemen aries of common libraries in many programming language the wheel.

Even so, r **basic linear algebra and multivariable ca**

W ll need to build custom implementations of ML al y need to adapt one to your tech stack or to expand its b e able to peel back ML algorithms and work with



R&D-Heav Positions

er roles r ML research and development. You may need to slate alg pers into working code. Or, you might research ancements based on your business's unique challenges.

p the academics. Learn practical ML from cticing professionals who will teach you **w to get real results with machine rning.**

<< Learn more about this 100% project-based machine learning masterclass

ther words, you'll be implementing algorithms from scratch much more often.

these positions, **mastery of both linear algebra and multivariable calculus** is a must.

The Best Way to Learn Math for Data Science

The self-starter way to learning math for data science is to **learn by "doing shit."** So we're going to tackle linear algebra and calculus by using them in real algorithms!

Even so, you'll want to learn or review the underlying theory up front. You don't need to read a whole textbook, but you'll want to learn the key concepts first.

Here are the 3 steps to learning the math required for data science and machine learning:

1. **Linear Algebra for Data Science**
   Matrix algebra and eigenvalues.
2. **Calculus for Data Science**
   Derivatives and gradients.
3. **Gradient Descent from Scratch**
   Implement a simple neural network from scratch.

Step 1: Linear Algebra for Data Science

Many machine learning concepts are tied to linear algebra. For example, PCA requires eigenvalues and regression requires matrix multiplication.

Also, most ML applications deal with high dimensional data (data with many variables). This type of data is best represented by matrices.

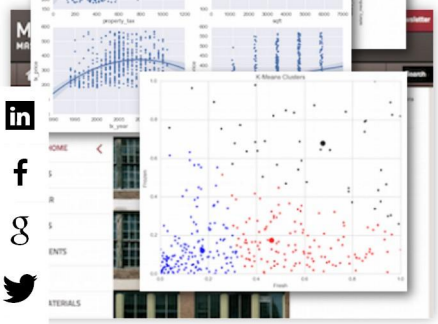
Here are a few of the best free resources we've found for learning linear algebra for data

science:



For application-heavy roles...

Khan Academy has short, practical linear algebra lessons. They cover the most important topics.



For R&D-heavy roles...

MIT OpenCourseWare offers a rigorous linear algebra class. The video lectures and course materials are all included.

p the academics. Learn practical ML from
cticing professionals who will teach you
w to get real results with machine
rning.
Learn more about this 100% project-based machine learning masterclass

If you only need to review:

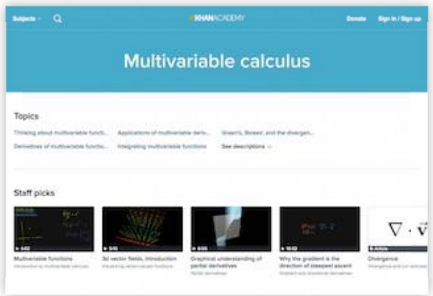
- Linear Algebra Review for Machine Learning (Video Series) - These are the optional linear algebra review videos for Andrew Ng's machine learning course. The entire 6-part series can be watched in under 1 hour. Recommended if you've taken linear algebra before and just need a quick review.
- The Matrix Cookbook (PDF) - Excellent reference resource for matrix algebra.

Step 2: Calculus for Data Science

Calculus is important for several key ML applications. For example. you'll need to be able to calculate derivatives and gradients for optimization.

In fact, one of the most common optimization techniques is gradient descent.

Here are some of the best resources for learning calculus for data science:



For application-heavy roles...

Khan Academy has short, practical multivariable calculus lessons. They cover the most important concepts.



For R&D-heavy roles...

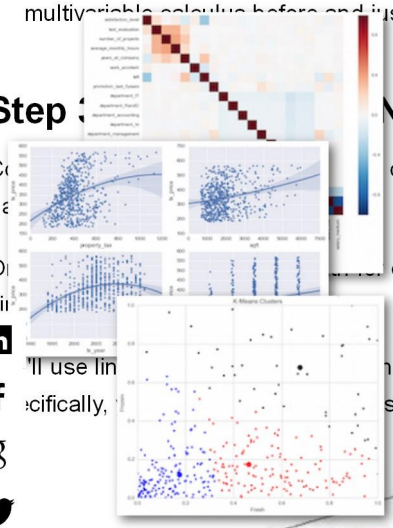
MIT OpenCourseWare offers a rigorous multivariable calculus class. The video lectures and course materials are all included.

And if you only need to review:

- Multivariable Calculus Review (Video) - This is quick review of multivariable calculus in the format of solving practice problems. Recommended if you've taken multivariable calculus before and just need a quick review.

How to Learn Practical Machine Learning in Just 1 Month...

Step 3: Build a Neural Network from Scratch



C out of the way. Now it's time for the really fun pa

O data science and machine learning is to build a sii

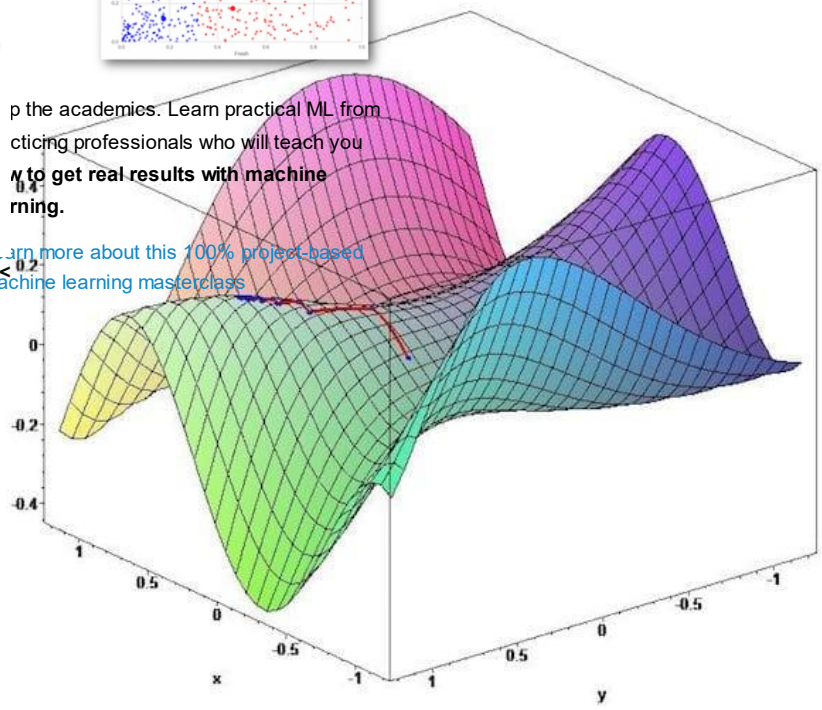
'll use lin ne network and calculus to optimize it.

cifically, scent from scratch.

p the academics. Learn practical ML from cticing professionals who will teach you **w to get real results with machine rning.**

Learn more about this 100% project-based machine learning masterclass

<<



Don't worry too much about the nuances of neural networks for now. It's ok if you're just following instructions and writing code. We'll cover machine learning in depth in another guide, as this is for targeted math practice.

Follow along with the tutorials, and review theory as you go along. Plus, you'll have a cool project to add to your portfolio afterward.

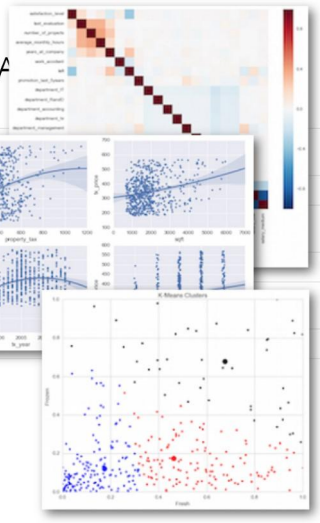
Here are a few awesome step-by-step guides:

- Neural Network in Python, Part 2 - This is an incredible tutorial that takes you through a simple neural network from end to end. It's packed with helpful illustrations, and you'll learn about how gradient descent fits in.
- Neural Nets to Recognize Handwritten Digits - We love this resource! This is a free online book that walks you through a famous application of neural networks. It explains ideas very intuitively, and it's the most in-depth tutorial in this list.
- Implementing a Neural Network from Scratch - A shorter tutorial that also takes you

through step-by-step.

How to Learn Practical Machine Learning in Just 1 Month

Share Google Linkedin Tweet



Leave A

*Name**

*Email**

Website

p the academics. Learn practical ML from
cticing professionals who will teach you
w to get real results with machine
rning.

bmit Comment

** Denotes Required Field*

Learn more about this 100% project-based machine learning masterclass

<<

Copyright © 2016 · EliteDataScience.com · All Rights Reserved